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CE205
Databases and Information Retrieval

Anomalies, Inference Rules of Functional Dependencies, Keys and Attribute Closure

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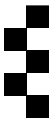
Lecturer

School of Computer Science and Electronic Engineering

Most of the slides based on

- *Database Systems: A Practical Approach to Design, Implementation, and Management* by Conoly and Begg, Global Edition/6th Edition.
- *Database Systems: Introduction to Databases to Data Warehouses* by Jukic, Vrbsky, Nestorov, sharma, 2nd Edition.

Keys in a Relation



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Keys and its Attributes

- A **key** is a combination of one or more columns that is used to identify particular rows in a relation.
- A relation may have more than one key, each is called a candidate key.
- **Primary key:** One of the candidate keys which is *arbitrarily selected* by the database designer.
- A **Prime attribute** are member of *some* candidate key
- A **Nonprime attribute** is not a prime attribute—that is, it is not a member of any candidate key.

Closure of an Attribute

- **The set of attributes that are functionally dependent on the attribute A is called Attribute Closure of A and it can be represented as A+.**

Computation of Attribute Closure

Problem: Compute the attribute closure of AB , i.e. $\{AB\}^+$,in relation R(ABCDE) with respect to the set of FDs :

$AB \rightarrow C$ (a)

$A \rightarrow D$ (b)

$D \rightarrow E$ (c)

$AC \rightarrow B$ (d)

Solution:

- Initially $\{AB\}^+ = \{AB\}$
- Using (a), $\{AB\}^+ = \{ABC\}$
- Using (b), $\{AB\}^+ = \{ABCD\}$
- Using (c), $\{AB\}^+ = \{ABCDE\}$

Candidate key and Superkey

- Both **candidate key** and **superkey** determine all of the other attributes in a relation.
- The only **difference** between them is that candidate key is a minimal superkey.
- In other words, **a proper subset** of the candidate key cannot determine all other attributes in a relation.

Finding Candidate Keys and Superkeys in a Relation using FDs (1)

Problem: Consider a relation R(A,B, C, D, E, F) With the following FDs: AB → C, C→ D, C→ E, E→ F, D→ A, C→B

Find all possible candidate keys?

Solution

- First rewrite all FDs such that right side contains only one attribute (given FDs are already in that form)
- Now find closure of {ABCDEF}, i.e. $\{ABCDEF\}^{+}$=?
- $\{ABCDEF\}^{+}$={ABCDEF} (using trivial rule of FD)
- Reduce attributes on left side, as AB→C, so remove C

Finding Candidate Keys and Superkeys in a Relation using FDs (2)

- $\{ABDEF\}^+=\{ABCDEF\}$
- Now using AB→C and C→D, remove D
- $\{ABEF\}^+=\{ABCDEF\}$
- Using AB→C and C→E, remove E
- $\{ABF\}^+=\{ABCDEF\}$
- Using AB→C, C→E and E→F, remove F, and we left with
- $\{AB\}^+=\{ABCDEF\}$
- As AB closure functionally determine set of all attributes of R, So AB is a superkey.

Finding Candidate Keys and Superkeys in a Relation using FDs (3)

- Moreover, no proper subset of AB can determine all attributes. Therefore, AB is a possible candidate key. So,
- Candidate key={AB}

- More candidate keys may also exist, if any attribute (A or B) is appearing on the right side of any given FD.
- A is on the right side of D→A, B is on the right side of C→B, so there is possibility of more candidate keys.
- First replace A with D in AB, we will get DB

Finding Candidate Keys and Superkeys in a Relation using FDs (4)

- Now determine closure of DB
- $\{DB\}^{+}=\{DBACEF\}$
- As DB also functionally determines all attributes, so it is a superkey.
- Moreover, no proper subset of DB functionally determines all attributes. Therefore, DB is a possible candidate key.
- So update possible candidate keys,

- Candidate keys={AB, DB}

Finding Candidate Keys and Superkeys in a Relation using FDs (5)

- Now replace B with C in AB and find closure
- $\{AC\}^+=\{ACBDEF\}$
- Although AC determines all attributes, its subset C can also find all attributes. Therefore, AC is not a candidate key.
- $\{C\}^+=\{CDEBFA\}$
- Therefore, C is a candidate key.
- So far, we found the following candidate keys: AB, DB and C.
- Prime attributes={C}

Candidate keys in Relation AD CAMPAIGN MIX

A	B	C	D	E	F	G	H	I	J
AdCampaignID	**AdCampaign Name**	**StartDate**	**Duration**	**Campaign MgrID**	**Campaign MgrName**	**ModusID**	**Media**	**Range**	**Budget Pctg**
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	1	TV	Local	50%
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	2	TV	National	50%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	1	TV	Local	60%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	3	Radio	Local	30%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	5	Print	Local	10%
333	FallBall20	6.9.2020	12 days	CM102	John	3	Radio	Local	80%
333	FallBall20	6.9.2020	12 days	CM102	John	4	Radio	National	20%
444	AutmnStyle20	6.9.2020	5 days	CM103	Nancy	6	Print	National	100%
555	AutmnColors20	6.9.2020	3 days	CM100	Roberta	3	Radio	Local	100%

Given FDs:

A→B
A→C
A→D
A→E
A→F
E→F
G→H
G→I
AG→J

Using the given FDs, find all possible candidate key in the relation?

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Final Remarks

Now you Know

- Update Operations
- Why redundant data is not good in a relation
- What are update anomalies
- How to find FDs in a relation
- How to find Candidate key(s) in a relation

For normalisation of a relation, still we need to know

- What are partial FD and Transitive FD
- Dependency Preserving Decomposition
- Lossless Join Decomposition

Final Destination

- 2NF, 3NF, BCNF, 4NF, 5NF

Suggested Readings

- D1: Database processing: fundamentals, design, and implementation (David M. Kroenke; David J. Auer) . 14th Edition (Global Edition). Pearson.
  - Chapter 3
- D2: The following sections from the textbook *Database Systems: A Practical Approach to Design, Implementation, and Management* by Conoly and Begg, Global Edition/6th Edition.
  - 14.2, 14.3, 14.4, 15.1
- https://opentextbc.ca/dbdesign01/chapter/chapter-11-functional-dependencies/